

**SafeResponse: A CENTRALIZED INCIDENT REPORTING AND AMBULANCE
DISPATCH PLATFORM OF MDRRMO FOR DISASTER AND MEDICAL
EMERGENCIES**

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A capstone project manuscript submitted to the faculty of the Department of Information Technology, Cavite State University, Naic Campus, Naic, Cavite in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology. Contribution No._____. Prepared under the supervision of Dr. Michelle C. Tanega

INTRODUCTION

In recent times, local governments in the Philippines have slowly moved into using digital tools for handling disasters. Because climate threats are getting stronger, quicker responses based on reliable information - alongside active community roles - are now essential. Data from the World Risk Index (2023) shows the country remains one of the world's three most at-risk nations; each year brings around twenty typhoons that result in massive economic losses. Even so, several town-level DRRM teams still use handwritten records, physical health forms - and outdated reporting styles - which delay actions while reducing live updates during crises.

Current emergency procedures struggle with multiple issues. Patient details are written by hand, which raises chances of mistakes, missing entries, or losing files

altogether. Reporting dangers usually happens after events occur, since MDRRMO units rely on scouts rather than live updates from locals.

In addition, hazard maps used by various LGUs don't update in real time, stay unchanged for long periods, or miss geographic tracking features, leaving blind spots when spotting new threats. Recent research shows automation helps record-keeping, yet many local teams lack proper support. Combining such tools could strengthen disaster planning, though few solutions fit small government units well. To tackle existing shortcomings, this research introduces SafeResponse, a digital platform designed for Naic's disaster response unit that supports emergency coordination alongside local hazard tracking.

The proposed SafeResponse system supports Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities by improving disaster preparedness, response, and resilience at the local level. It provides a centralized and real-time platform for hazard monitoring, incident reporting, and emergency coordination, enabling faster and more informed decision-making. Features such as Real-Time Geo-Referenced Hazard Mapping, Crowdsourced Hazard Reporting, and Centralized Dashboard & Analytics enhance situational awareness and help build safer, disaster-resilient communities. In addition, the system supports Sustainable Development Goal (SDG) 13: Climate Action by strengthening local capacity to respond to climate-related hazards like typhoons and floods. Through Digital Records & Data Archiving and Automated Report Summarization, it enables better analysis of past incidents for data-driven disaster planning. Additionally, Crowdsourced Hazard Reporting promotes community participation, supporting proactive disaster management and a more climate-resilient local government unit.

Project Context

The Municipality of Naic, Cavite is frequently exposed to natural hazards such as typhoons, flooding, and other climate-related emergencies, requiring an efficient

and well-coordinated disaster response system. The Municipal Disaster Risk Reduction and Management Office (MDRRMO) plays a critical role in handling emergency incidents, coordinating responders, and ensuring public safety. However, many of its current processes rely on manual documentation, fragmented reporting methods, and static hazard mapping, which can delay decision-making and limit real-time situational awareness during emergencies.

In this context, there is a growing need for a centralized and technology-driven solution that can support faster, more accurate, and data-driven disaster response operations. The lack of integrated systems for incident reporting, patient data management, hazard monitoring, and resource coordination highlights the importance of digital transformation within local government disaster management. To address these challenges, this study introduces SafeResponse: A Centralized Incident Reporting and Ambulance Dispatch Platform of MDRRMO for Disaster and Medical Emergencies.

. The system is designed to modernize disaster response operations by integrating key features such as Automated Report Summarization, Digital Patient Data Collection, Crowdsourced Hazard Reporting, and Real-Time Geo-Referenced Hazard Mapping into a single platform. Additionally, it includes Incident & Dispatch Management, Centralized Dashboard & Analytics, Digital Records & Data Archiving, and User Access Levels & Security to ensure efficient coordination, secure data handling, and improved decision-making.

By implementing SafeResponse, the Naic MDRRMO can enhance its operational efficiency, improve communication among responders and the community, and strengthen its capability to respond to emergencies effectively. The project aims to contribute to a more resilient, informed, and disaster-ready municipality through the use of integrated digital technologies.

Objectives of the Study

The general objective of the study is to develop and implement SafeResponse: A Centralized Incident Reporting and Ambulance Dispatch Platform of MDRRMO for Disaster and Medical Emergencies that can be utilized by the Naic Municipal Disaster Risk Reduction and Management Office (MDRRMO) to improve operational efficiency, enhance real-time situational awareness, and streamline data handling during emergencies.

Specifically, the study aims to:

1. Design the SafeResponse system architecture that:
 - a. Integrates Automated Report Summarization capabilities for quick review of incident, ambulance, dispatch, and medical narratives;
 - b. Incorporates a structured interface for Digital Patient Data Collection (including vitals, treatments, ambulance response details, and time logs) for field responders;
 - c. Provides a mechanism for Crowdsourced Hazard Reporting via web interfaces to capture real-time community information; and
 - d. Displays a Real-Time Geo-Referenced Hazard Map that combines both official and crowdsourced incident data.
 - e. Includes an Incident Report and Dispatch Management module for monitoring emergency calls, ambulance dispatch, responder coordination, and incident status updates.
2. Develop the SafeResponse system using the following:
 - a. Javascript React for web application development and system logic;
 - b. MySQL and PostgreSQL for robust database management and geospatial data storage;
 - c. HTML5, CSS3, JavaScript, and relevant mapping APIs (e.g., Leaflet) for the user interface and mapping functions;

- d. Web Browsers (for the MDRRMO Command Center) for accessing the system; and
 - e. Test the system in terms of unit, integration, and user acceptance testing.
3. Evaluate the system using the adapted ISO 25010 evaluation instrument focusing on functionality, performance efficiency, and usability.
 4. Prepare an implementation plan detailing the deployment, training, and long-term maintenance strategy for the Naic MDRRMO.

Purpose and Description

The purpose of this study is to develop SafeResponse: A Centralized Incident Reporting and Ambulance Dispatch Platform of MDRRMO for Disaster and Medical Emergencies, a web-based platform designed to support the Naic MDRRMO in improving disaster response, coordination, and hazard monitoring. The system addresses challenges such as manual reporting, delayed information processing, and limited real-time hazard visibility by providing a centralized, secure, and user-friendly solution.

The SafeResponse system serves as a centralized platform for managing disaster-related information, including real-time incident tracking, hazard mapping, and emergency coordination. It features a Centralized Dashboard for monitoring active incidents and supporting faster, data-driven decision-making. To improve efficiency, the system includes Digital Patient Data Collection for accurate medical documentation and Automated Report Summarization for quick analysis of reports.

For community involvement, the system provides Crowdsourced Hazard Reporting, allowing residents to submit hazards with photos and location data. These reports are integrated into the Real-Time Geo-Referenced Hazard Mapping, which

visualizes incidents and hazard levels for better situational awareness. Additionally, the system includes Incident & Dispatch Management, Digital Records & Data Archiving, and User Access Levels & Security to ensure efficient operations and secure data handling.

Overall, the SafeResponse system aims to enhance disaster preparedness, improve response efficiency, and support a more organized and data-driven approach to emergency management in Naic.

Time and Place of the Study

This study is being conducted in the Municipality of Naic, Cavite, Philippines. The primary setting of the study is the Municipal Disaster Risk Reduction and Management Office (MDRRMO) of Naic. From December 2025 to May 2026.

Scope

The scope of this study covers the development of SafeResponse including administrators, staff, responders, and registered community users. The system focuses on improving disaster preparedness, emergency response coordination, hazard monitoring, and incident management through a centralized digital platform. It includes features such as Automated Report Summarization, Digital Patient Data Collection, Crowdsourced Hazard Reporting, Real-Time Geo-Referenced Hazard Mapping, Incident & Dispatch Management, Centralized Dashboard & Analytics, Digital Records & Data Archiving, and User Access Levels & Security. The system also supports community participation by allowing users to submit hazard reports with descriptions, photos, and GPS-based locations to help MDRRMO gather real-time disaster-related information.

Limitation

The proposed SafeResponse system depends on a stable internet connection for real-time synchronization, hazard updates, and cloud-based database access. Offline reporting and offline map synchronization are not fully supported in the current version of the system. The study is limited to the Municipality of Naic and is not intended for nationwide MDRRMO deployment.

Furthermore, the system is only accessible through web browsers and does not include a standalone mobile application. Integration with advanced external emergency systems, weather forecasting APIs, SMS gateways, and national disaster databases is beyond the scope of the study. In addition, the Automated Report Summarization feature is limited to summarizing submitted textual reports and may still require human verification for critical emergency decisions. The accuracy of crowdsourced hazard reports also depends on the reliability and correctness of information submitted by users.

Lastly, the study mainly focuses on system development and functionality evaluation and does not include large-scale operational deployment or long-term field performance assessment, IoT sensors, drones, CCTV systems, shortage of emergency vehicles and utility vehicles, in cases where the patient requires medical attention beyond the capacity of the municipality of Naic, Cavite and also proper use of the mobile application by community users and responders.

Conceptual Framework

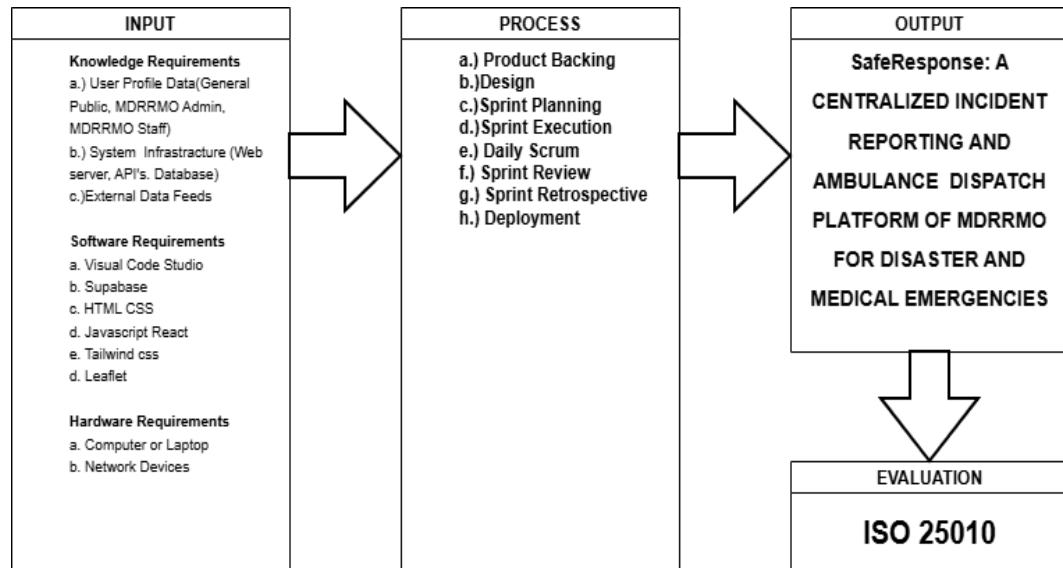


Figure 1. Conceptual Framework of SafeResponse: A Centralized Incident Reporting and Ambulance Dispatch Platform of MDRRMO for Disaster and Medical Emergencies

Definition of Terms

To better understand the study, the following terms are operationally defined:

Automated Report Summarization refers to the system feature that automatically generates concise summaries of incident and medical reports to help MDRRMO personnel quickly review important details and improve documentation efficiency.

Centralized Dashboard & Analytics refers to the monitoring interface that provides real-time visualization of active incidents, hazard reports, heatmaps, trends, and historical disaster data to support data-driven decision-making.

Cloud-Based System refers to the online infrastructure used by the SafeResponse system to store, manage, and synchronize disaster-related data through internet connectivity, allowing authorized users to access information remotely.

Crowdsourced Hazard Reporting refers to the feature that allows community members to submit hazard reports, including photos, descriptions, and GPS-based locations, through web or mobile platforms.

Digital Patient Data Collection refers to the electronic recording of patient information such as vital signs, injuries, treatments, responder details, and time logs during emergency response operations.

Digital Records & Data Archiving refers to the system module responsible for storing, organizing, and retrieving incident reports, patient records, hazard logs, and other disaster-related documents in digital form.

Geo-Referenced Hazard Mapping refers to the process of displaying hazards and incidents on a digital map using geographic coordinates and location-based data for real-time situational awareness.

Incident & Dispatch Management refers to the feature used to log incidents, assign responders, monitor response progress, and track emergency operations from report submission to resolution.

ISO 25010 refers to the international software quality standard used to evaluate the functionality, usability, reliability, security, and performance efficiency of the developed system.

Real-Time Monitoring refers to the live updating capability of the system that allows users and MDRRMO personnel to immediately view new incidents, hazard reports, and response activities.

REVIEW OF THE RELATED LITERATURE

This chapter presents the related literature and studies that helped the researchers understand the concepts, technologies, and systems related to disaster management, hazard mapping, emergency response, and web-based reporting systems.

Related Literature

Emergency Ambulance Dispatch Systems. Hua and Zaman (2020) discussed the importance of optimized ambulance dispatch systems in improving emergency response operations. The study emphasized that efficient dispatch and resource allocation can help reduce response time and improve coordination during emergency situations. This literature supports the proposed SafeResponse system, particularly its Incident Report and Dispatch Management feature, which is designed to monitor emergency incidents, ambulance dispatch, responder coordination, and incident status updates for faster and more organized disaster response operations.

Multi-Hazard Prediction in Disaster Management. Multi-hazard disaster prediction models improve the accuracy of damage assessment and emergency planning. Salvaña (2025) proposed the Multi-Hazard Bayesian Hierarchical Model (MH-BHM), which analyzes the interaction of multiple hazards using historical Philippine tropical-cyclone data. The study emphasized that integrating multiple disaster factors results in more reliable risk assessment and forecasting. This literature supports the proposed SafeResponse system, particularly its Real-Time Geo-Referenced Hazard Mapping and Centralized Dashboard & Analytics, which are designed to monitor and visualize multiple hazards for better disaster response operations.

Disaster Preparedness and Human Factors. Human preparedness and decision-making are essential factors in effective disaster response. Gumasing et al. (2022) examined the factors affecting perceived preparedness during super typhoons using an ergonomics-based approach. Their findings highlighted the importance of information accessibility, communication, and system usability in disaster preparedness. This literature supports the development of the SafeResponse system by emphasizing the need for a user-friendly and accessible platform that improves coordination, situational awareness, and emergency response efficiency.

Local Government Disaster Risk Reduction Management. Local Government Units (LGUs) continue to face challenges in disaster preparedness, response coordination, and recovery operations. Tabao (2025) assessed the effectiveness of disaster risk reduction management strategies among Philippine LGUs and found that many municipalities experience poor coordination, delayed reporting, and insufficient technological support. The study emphasized the importance of integrated digital systems for real-time hazard monitoring and incident management. This literature supports the proposed SafeResponse system because its centralized platform aims to improve coordination, reporting, and disaster response operations within the Naic MDRMO.

Artificial Intelligence in Disaster Response. Artificial Intelligence (AI) has become increasingly important in improving disaster prediction, mitigation, and emergency response systems. Baltazar, Florencio, Vicente, and Belizario (2024) explained that AI-driven technologies enhance forecasting accuracy and hazard analysis by processing large amounts of real-time and historical data. The study emphasized the importance of structured and integrated digital records for effective disaster management. This literature supports the SafeResponse system, particularly its Automated Report Summarization, Crowdsourced Hazard Reporting, and Digital

Records & Data Archiving features, which provide organized and accessible disaster-related information.

Crowdsourcing and Task Coordination System. Crowdsourcing platforms improve collaboration, task assignment, and information gathering in digital systems. Zhen et al. (2021) conducted a systematic literature review on crowdsourcing usage, task assignment methods, and platform integration. Their study highlighted that effective coordination and real-time information sharing are essential for improving operational efficiency and timely response. This literature supports the proposed SafeResponse system because its Crowdsourced Hazard Reporting and Incident & Dispatch Management features are designed to improve communication, coordination, and resource allocation during emergencies.

Smart Systems for Disaster Risk Reduction. Smart systems and emerging technologies improve disaster risk reduction and emergency response operations. Dixit (2025) discussed the application of smart systems in disaster management and emphasized the need for integrated platforms capable of real-time hazard monitoring and coordinated response. The study found that centralized digital systems improve situational awareness and support faster decision-making during disasters. This literature supports the proposed SafeResponse system because its centralized dashboard and real-time hazard mapping features are intended to strengthen disaster response coordination within the Naic MDRMO.

Mobile-Based Crowdsourcing Platforms for Disaster Response. Mobile-based crowdsourcing platforms enable communities to actively participate in disaster reporting and emergency response. Lacasandile, Gunay, and Bonifacio (2022) developed E-Saklolo, a mobile-based crowdsourcing platform for disaster risk management in the Philippines. Their findings revealed that community-driven

reporting improves the speed of information sharing and enhances emergency response coordination. This literature supports the proposed SafeResponse system because its Crowdsourced Hazard Reporting and mobile accessibility features encourage community participation and real-time hazard reporting.

Geospatial Technology in Disaster Preparedness. Geospatial technologies play an important role in improving disaster preparedness and hazard monitoring. Mongabay (2021) reported that the Philippines has been improving disaster preparedness through the use of geospatial technologies for mapping and monitoring hazards. The article emphasized that real-time geospatial information enhances disaster awareness and emergency planning. This literature supports the Real-Time Geo-Referenced Hazard Mapping feature of the SafeResponse system, which aims to provide accurate and updated hazard visualization for MDRRMO personnel and the community.

Digital Preservation of Government Records. Digital systems are essential for preserving government records and ensuring long-term access to important information. Diaz Calderón et al. (2025) emphasized the importance of smart digital systems for preserving government records and preventing data loss during emergencies. The study highlighted that electronic record management improves accessibility, organization, and data security. This literature supports the Digital Records & Data Archiving feature of the proposed SafeResponse system, which is designed to securely store and manage disaster-related reports and records.

Role-Based Access Control Systems. Role-based access control systems help ensure data privacy, security, and proper information management in digital platforms. Idais (2022) developed a visual tool for defining role-based access control systems and emphasized the importance of restricting access to sensitive information based on user roles. The study highlighted that secure access

management improves accountability and protects confidential data. This literature supports the User Access Levels & Security feature of the SafeResponse system, which ensures that only authorized personnel can access and manage disaster-related information.

JavaScript Frameworks for Web Applications. Satrom (2018) discussed the importance of choosing the appropriate JavaScript framework for web application development. The study emphasized that modern JavaScript frameworks improve system performance, user interface responsiveness, scalability, and development efficiency. This literature supports the use of React and JavaScript in the proposed system to create an interactive, responsive, and user-friendly platform for disaster response management and hazard mapping.

Backend-as-a-Service (BaaS) Platforms. Lorenz (2024) discussed the advantages of using Supabase as a Backend-as-a-Service (BaaS) platform for modern web applications. The study highlighted features such as real-time database synchronization, authentication, cloud-based storage, and scalability, which simplify backend development and improve system efficiency. This literature supports the use of Supabase in the proposed system for secure cloud-based database management, real-time data processing, and efficient handling of disaster-related information and user authentication.

Systems Technical Background

The proposed system is developed as a responsive web-based application that can be accessed through desktop and mobile web browsers. The researchers used HTML, CSS, and JavaScript for developing the client-side interface to ensure that the system is interactive, responsive, and user-friendly. In addition, the React JavaScript library was utilized to create dynamic user

interfaces and improve the responsiveness of the platform, particularly for dashboards, hazard maps, and real-time data updates.

For the server-side development, the researchers used Node.js to handle backend processes, real-time communication, and system functionalities such as incident management, notifications, and report processing. Node.js also supports efficient handling of multiple user requests during emergency operations.

The system uses PostgreSQL as its primary relational database management system for securely storing and managing disaster-related information, including incident reports, patient records, hazard reports, operational logs, and user accounts. To support cloud-based services, authentication, and database synchronization, the researchers integrated Supabase Backend-as-a-Service (BaaS) into the system architecture.

For hazard visualization and geographic monitoring, the researchers used Leaflet, an open-source JavaScript mapping library that enables real-time geo-referenced hazard mapping and interactive map displays. This technology allows the system to present hazard locations, incident markers, and community-submitted reports effectively.

The development and coding environment utilized Visual Studio Code (VSC) for system programming, debugging, and project management. The researchers also used draw.io for creating system diagrams such as the Use Case Diagram, Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), and System Architecture Diagram.

Overall, these technologies and tools were selected to ensure that the SafeResponse system is secure, scalable, responsive, and capable of supporting

efficient disaster response and hazard management operations for Naic MDRRMO.

Table 1 summarizes an enumeration of the proposed SafeResponse system and research of existing other studies or platforms that is nearest to it based on modules and functional requirements.

Table 1. Comparison table of related studies

MODULES / FEATURES	S1	S2	S3	S4	PROPOSED STUDY (SafeResponse)
Ambulance Dispatch & Responder Coordination		✓		✓	✓
Automated Report Summarization	✓		✓		✓

Digital Patient Data Collection		✓	✓		✓
Crowdsourced Hazard Reporting	✓	✓		✓	✓
Real-Time Geo-Referenced Hazard Mapping	✓		✓	✓	✓
Incident Dispatch & Management		✓	✓	✓	✓
Centralized Dashboard & Analytics	✓		✓		✓

Digital Records & Data Archiving			✓	✓	✓
User Access Levels & Security	✓			✓	✓
METHODOLOGY	AI-Based	Crowdsourcing	GIS-Based	Community-Based	Scrum Methodology

Legend:

S1 – Salvaña et al. (2025) / Baltazar et al. (2024)

S2 – Gumasing et al. (2022) / Zhen et al. (2021) / Hua and Zaman (2020)

S3 – Lacasandile et al. (2022) / Mongabay (2021)

S4 – Tabao (2025) / Dixit (2025) / Idais (2022)

Synthesis

The reviewed literature and related studies provide the theoretical and technical foundation for the development of the SafeResponse system. The studies highlight the importance of real-time disaster monitoring, integrated emergency response systems, community participation, secure digital records management, and data-driven decision-making in improving disaster preparedness and response operations.

Salvaña (2025) and Baltazar et al. (2024) emphasized the importance of multi-hazard prediction models, Artificial Intelligence, and structured digital data in improving disaster forecasting and emergency response. These concepts support the proposed system's features such as Real-Time Geo-Referenced Hazard Mapping, Automated Report Summarization, and Centralized Dashboard & Analytics. In addition, Hua and Zaman (2020) highlighted the significance of technology-driven disaster management systems in improving coordination, communication, and decision-making during emergencies.

Moreover, Tabao (2025) and Dixit (2025) identified the challenges faced by Local Government Units due to fragmented and manual disaster management systems. The proposed system addresses these issues through Incident & Dispatch Management, Digital Records & Data Archiving, and Crowdsourced Hazard Reporting integrated into a centralized platform for Naic MDRRMO.

In addition, Lacasandile et al. (2022) and Zhen et al. (2021) highlighted the value of crowdsourcing and real-time information sharing in disaster response. Meanwhile, Diaz Calderón et al. (2025) and Idais (2022) emphasized the need for secure digital records and role-based access control, which are supported in the proposed system through PostgreSQL, Supabase BaaS, and User Access Levels & Security.

Furthermore, Satrom (2018) stressed the importance of modern JavaScript frameworks in developing responsive and scalable web applications, supporting the use of React and JavaScript in the system. Lorenz (2024) also discussed the advantages of Supabase for real-time database synchronization, authentication, and cloud-based data management.

Overall, the proposed SafeResponse system responds to the gaps identified in previous studies by combining real-time hazard monitoring, digital reporting, crowdsourced participation, secure cloud-based data management, and centralized emergency coordination into one integrated disaster management platform.

METHODOLOGY

This chapter includes the design of software, system, product, or process, system development, system testing and evaluation, data analysis and implementation plan of the study.

Design of Software, System, Product and/or Process

The design of the proposed SafeResponse system focuses on improving disaster response management, hazard monitoring, and emergency coordination within the Municipality of Naic. The system provides features such as Automated Report Summarization, Digital Patient Data Collection, Crowdsourced Hazard Reporting, Real-Time Geo-Referenced Hazard Mapping, Incident & Dispatch Management, and Centralized Dashboard & Analytics. These features help MDRRMO personnel manage incidents, monitor hazards, and respond to emergencies more efficiently and accurately.

The system also includes role-based access control to ensure that only authorized personnel can access and manage sensitive disaster-related information. The interface was designed to be simple, responsive, and user-friendly so that both MDRRMO personnel and community users can easily navigate and use the system through web browsers. In addition, the system supports cloud-based data management and digital records archiving to improve data accessibility, organization, and security.

Overall, the design integrates real-time monitoring, digital reporting, and centralized emergency management into a single web-based platform to support efficient disaster preparedness and response operations.

Figure 2 presents the HIPO diagram of the proposed SafeResponse system and illustrates the major modules and processes of the platform. The system is divided into key functions including Hazard Mapping, Incident Reporting, Outpatient Check-Up Service, Borrowing Vehicle, and Appointment Booking. Each module is connected to supporting processes such as User Data Management, Patient Data Recording, Incident Report Processing, Report Generation, Hazard Map Visualization, Appointment and Vehicle Approval, and Inventory Management. The diagram shows the overall structure and flow of the system, making the functions and interactions of each module easier to understand.

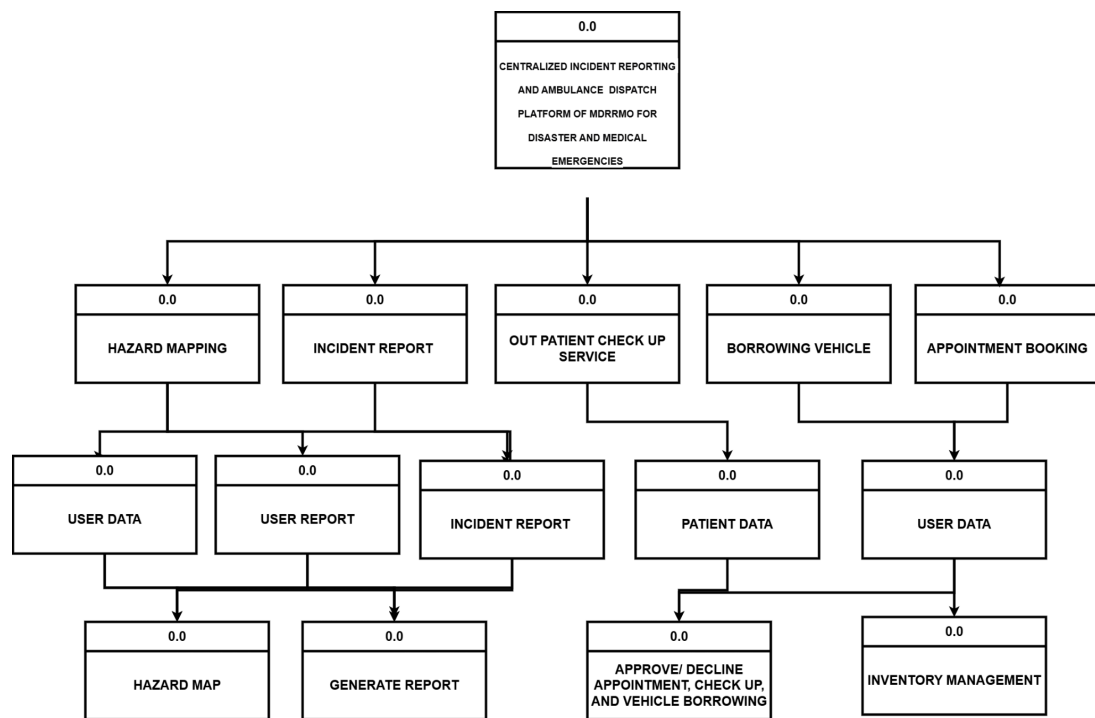


Figure 2. HIPO diagram of SafeResponse

Figure 3 presents the use case diagram of the proposed SafeResponse system. It illustrates the interactions between the four main actors of the system which are the Guest, Registered User, Staff, and Admin. Each actor has specific functions and access levels based on their role in the emergency response and community hazard mapping process.

The Guest can access public features of the system such as viewing the homepage, hazard map, weekly incident reports, and about page. Guests may also report incidents and hazards, borrow vehicles, and book appointments. The Registered User has additional capabilities, including filling out OPD check-up forms, tracking appointments, viewing notifications, and accessing incident reports.

The Staff is responsible for operational tasks within the system. Their functions include managing inventory, approving users, processing appointments, handling vehicle requests, and monitoring hazard and incident reports.

Meanwhile, the Admin has the highest level of access and oversees the overall management of the system. The Admin can approve hazard reports and appointments, generate reports, manage users and inventory, and monitor all incident-related activities to ensure efficient emergency response operations.

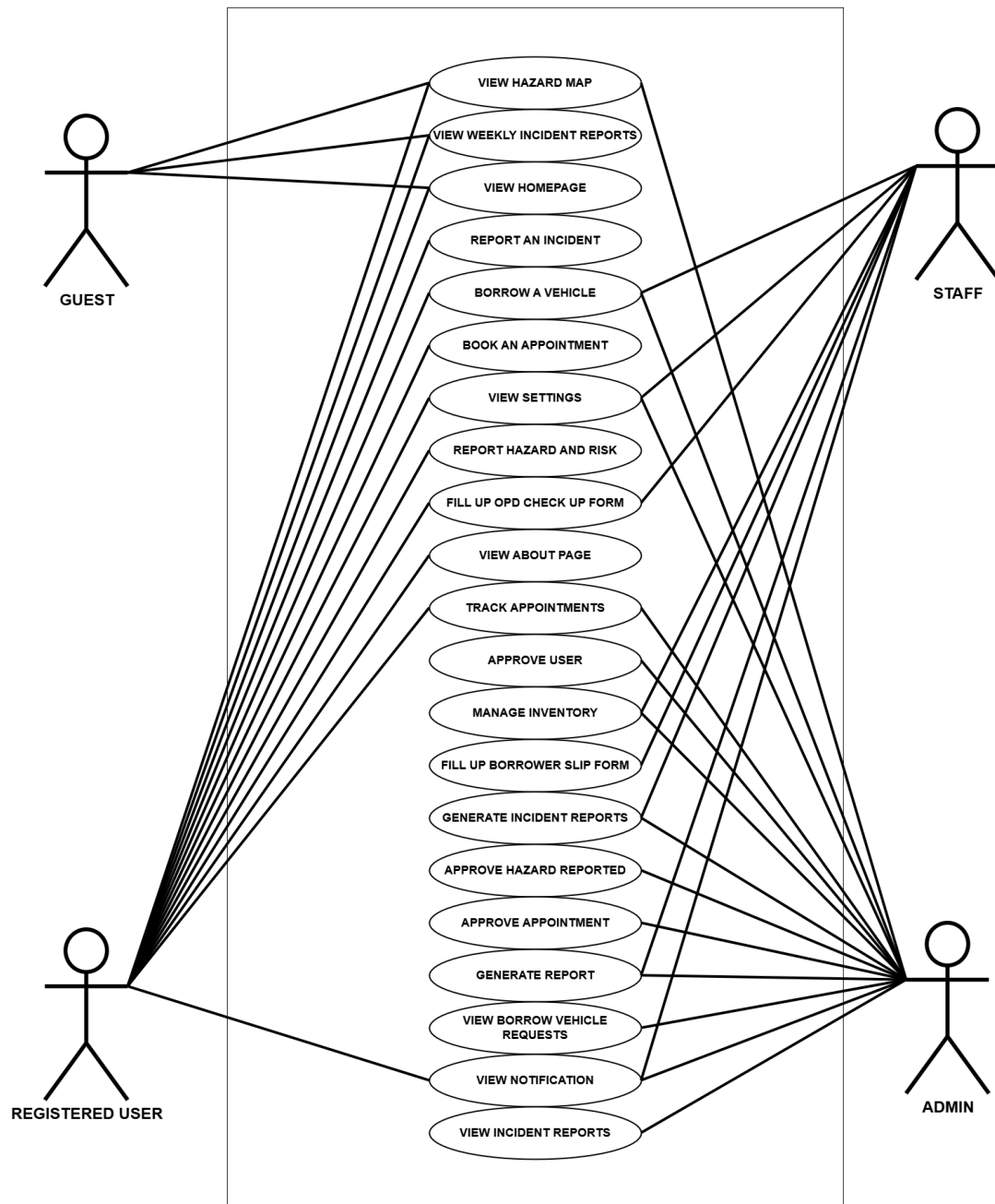


Figure 3. Use case diagram of SafeResponse

Entity Relationship Diagram or ERD of the eMDRMMO briefly shows the connection and communication of the tables inside the system's database figure 4 shows the detailed explanation of relationships between database's tables

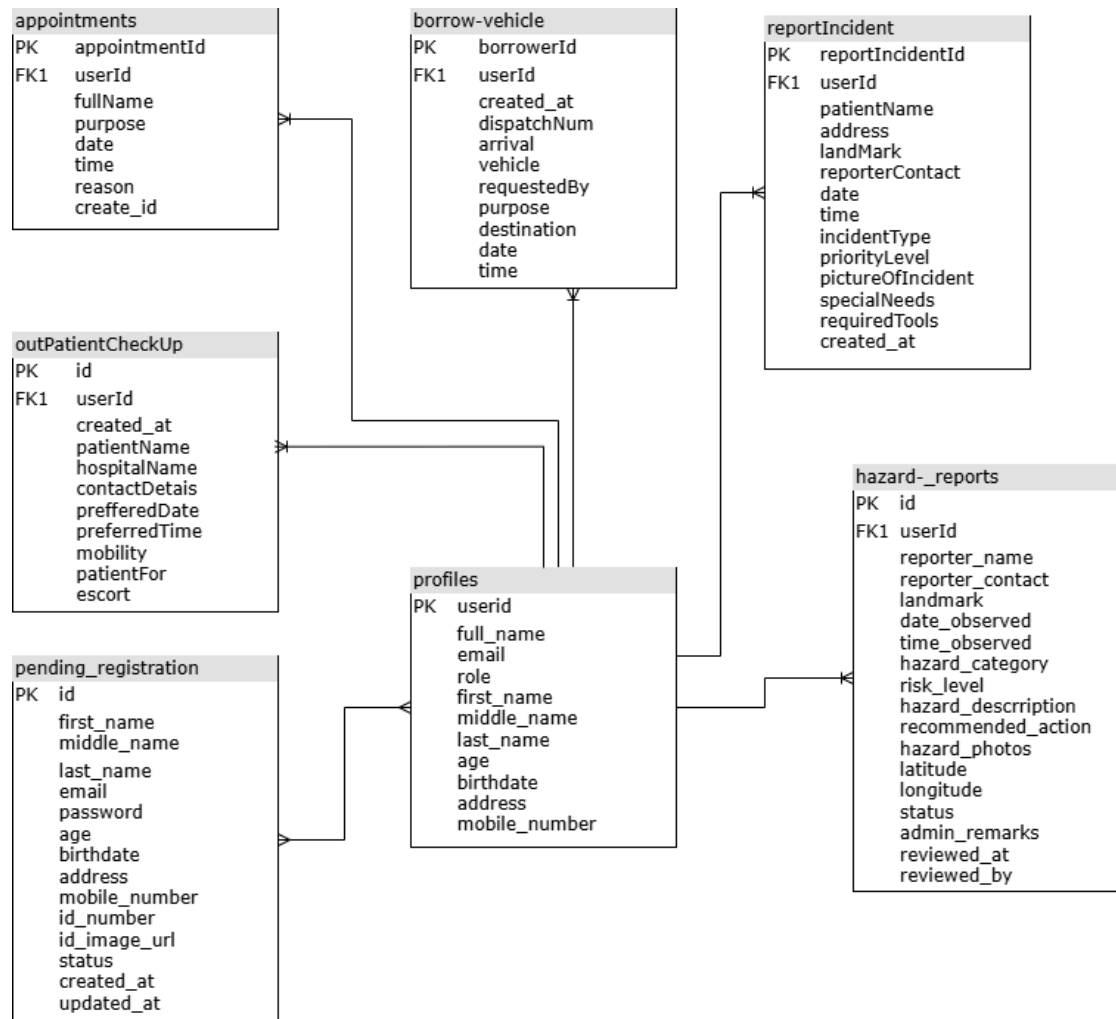


Figure 4. Entity Relationship Diagram of SafeResponse

Figure 5 shows the Fishbone Diagram of the SafeResponse. This discusses the possible barriers while conducting the research. It consists of 4Ms which are Manpower, Method, Machine and Materials.

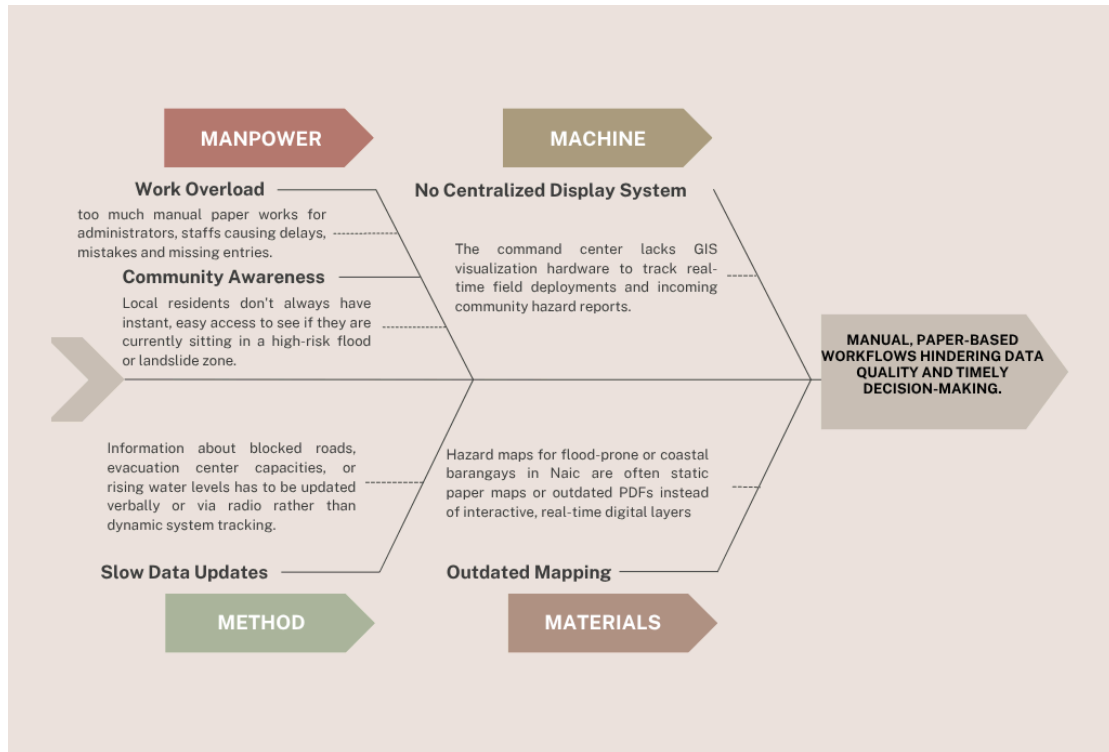


Figure 5. Fishbone Diagram of SafeResponse

The Data Flow Diagram of the SafeResponse was shown in figure 6. This shows the pathway of the data, How and where the commands go inside the system.

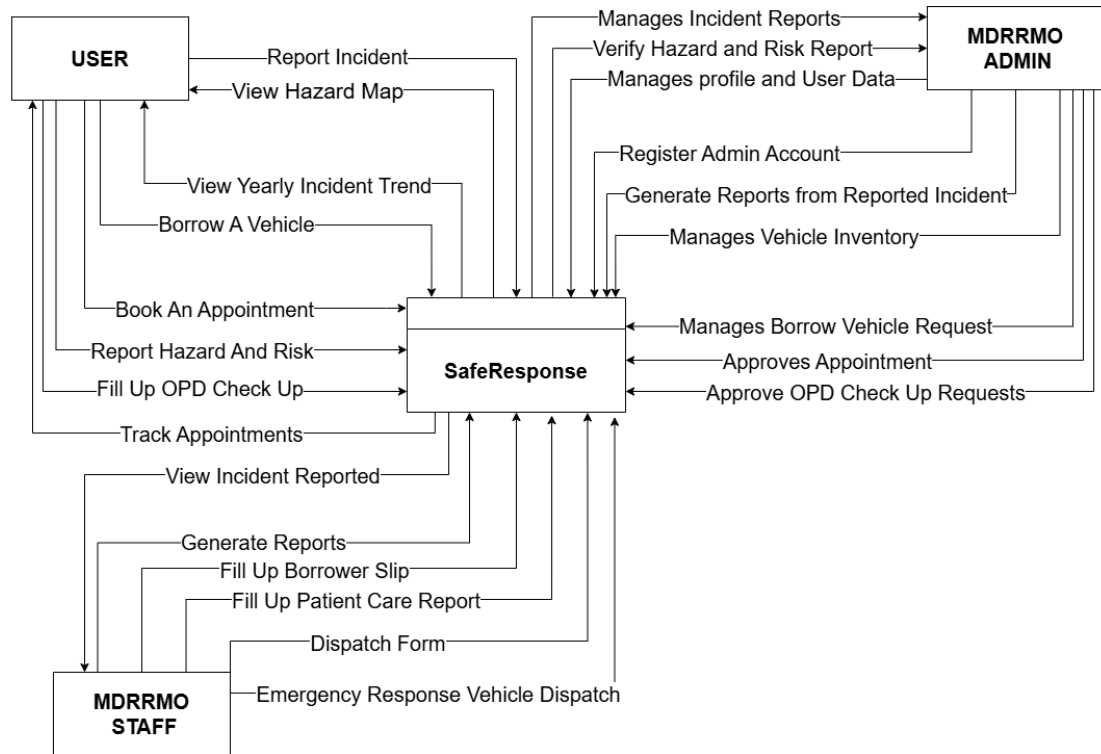


Figure 6. Context Flow Diagram of SafeResponse

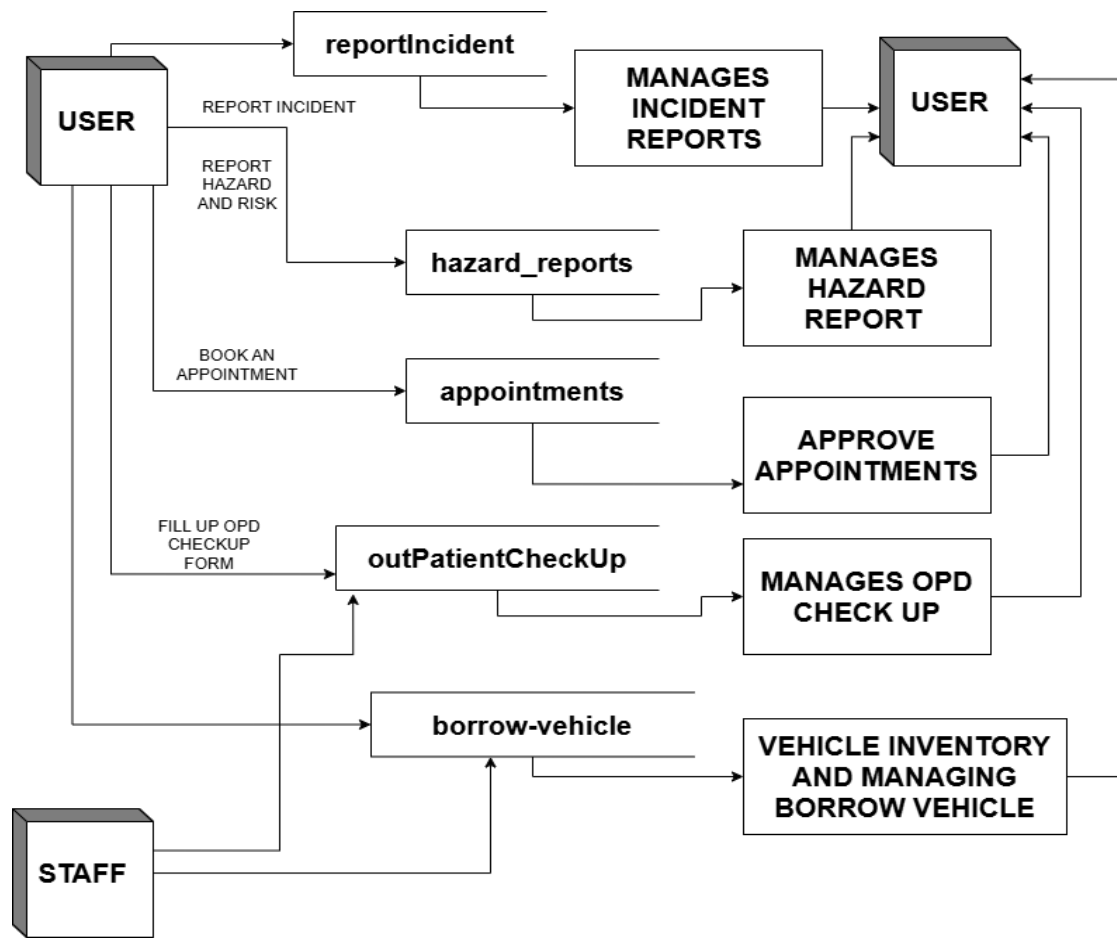


Figure 7. Dataflow Diagram of the SafeResponse

Table 2 presents the technical resources required for the implementation and operation of the proposed SafeResponse system. These resources include computers or laptops, internet connection, web browsers, database servers, and cloud-based services needed to support the system's functionalities such as hazard reporting, incident management, appointments, outpatient check-up records, and vehicle borrowing management. Since the system is web-based, users and MDRMO personnel can access the platform using any device with a supported web browser and internet connection. The system also utilizes PostgreSQL and Supabase for secure cloud-based database management, while Leaflet is used for hazard mapping and location visualization. In addition, Visual Studio Code serves as the development environment used by the researchers during the development of the

system. These resources ensure that the proposed system can effectively support the disaster response and management operations of the Naic MDRRMO.

Table 2. List of technical resources available in the organization

RESOURCES	QUANTITY
Naic MDRRMO Admin	1 per user
Naic MDRRMO Staff	1 per user
User (General Public)	1 per user

Table 3. Presents the estimated annual cost of the project. Resources such as domain name, web hosting and taxes

Table 3. Estimated cost of project

RESOURCES	COST
Hostinger Hosting Plan	ongoing
Domain Name Purchase	ongoing
VAT/Taxes	ongoing
Total Estimated Annual Cost	

Requirements analysis.

The functional requirements define the core services and operations that the system must provide. These include secure user authentication and role-based access control for Admin Office, Operational Team/All Staff, General Public Users, and Guests. The system must support Automated Report Summarization for faster review of incident and medical reports, Digital Patient Data Collection for recording emergency medical information, Crowdsourced

Hazard Reporting for community-submitted hazard data, and Real-Time Geo-Referenced Hazard Mapping using Leaflet technology. In addition, the system must provide Incident and Dispatch Management, Centralized Dashboard and Analytics, Digital Records and Data Archiving, and real-time notifications to improve emergency response operations and decision-making.

On the other hand, the non-functional requirements focus on the quality and operational standards of the system. The proposed SafeResponse system must maintain real-time performance, system reliability, and secure cloud-based data storage using PostgreSQL and Supabase. The system should provide a user-friendly and responsive interface accessible through desktop and mobile web browsers. It must also ensure data security, maintainability, scalability, portability, and availability to support continuous disaster response operations and future system improvements.

Software Design

The design follows the software quality characteristics defined in the ISO/IEC 25010 Software Quality Model. These characteristics include functional suitability, reliability, usability, performance efficiency, security, maintainability, and portability. Applying these quality standards ensures that the system operates efficiently and provides reliable support for disaster response, hazard monitoring, and emergency management operations.

For the system design process, the researchers utilized draw.io to create wireframes, system architecture diagrams, Entity Relationship Diagrams (ERD), Data Flow Diagrams (DFD), and interface layouts. These design tools helped visualize the structure, workflow, and user interaction of the system to ensure that the platform

would be organized, responsive, and easy to use for MDRRMO personnel and community users.

The system was designed using a Three-Tier Web Architecture composed of the Presentation Layer, Application Layer, and Data Layer. The frontend interface was developed using HTML, CSS, JavaScript, and React, while the backend logic and real-time functionalities were supported using Node.js. For database and cloud services, the system utilized PostgreSQL integrated with Supabase Backend-as-a-Service (BaaS) to securely store and manage disaster-related information.

The prototypes and system interfaces were evaluated through consultations, testing, and feedback sessions under the supervision of the Department Head of Naic MDRRMO. Through continuous refinement and feedback integration, the researchers improved the system's functionality, usability, and overall performance to ensure that the proposed SafeResponse platform becomes a reliable, accessible, and user-friendly disaster management solution for the Municipality of Naic.

System Development

The development of SafeResponse: A Centralized Incident Reporting and Ambulance Dispatch Platform of MDRRMO for Disaster and Medical Emergencies adapted the Agile Scrum model with seven stages namely...

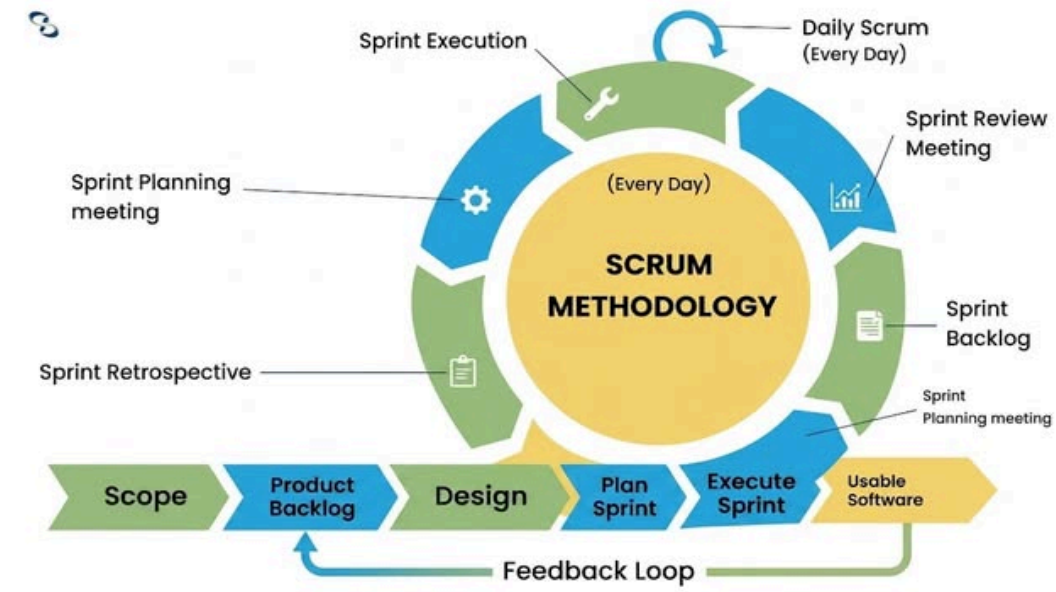


Figure 8. Scrum Methodology

The researchers utilized the Scrum methodology in developing the proposed SafeResponse system. Scrum allowed the researchers to divide the project into manageable tasks called sprints, enabling continuous development, testing, and refinement of system features based on the needs and feedback of Naic MDRRMO personnel. The Scrum methodology consists of several stages, namely Product Backlog, Sprint Planning, Sprint Execution, Daily Scrum, Sprint Review, Sprint Retrospective and Deployment.

Product Backlog. During this phase, the researchers gathered and listed all system requirements through interviews, consultations, and observations with Naic MDRRMO personnel. Existing issues such as manual incident reporting, delayed hazard updates, paper-based patient records, and fragmented communication were identified. Based on these findings, the researchers defined the major system features including Automated Report Summarization, Digital Patient Data Collection, Crowdsourced Hazard Reporting, Real-Time Geo-Referenced Hazard Mapping, Incident & Dispatch Management, and Centralized Dashboard & Analytics.

Sprint Planning. In this stage, the researchers organized and prioritized the identified requirements into smaller and manageable development tasks for each sprint cycle. The researchers planned the objectives, features, and functionalities to be completed within a specific development period to ensure organized and efficient progress throughout the project.

Sprint Execution. During sprint execution, the researchers designed and developed the system incrementally using Visual Studio Code (VSC). The frontend was developed using HTML, CSS, JavaScript, and React, while Node.js handled backend processing and real-time communication. PostgreSQL integrated with Supabase Backend-as-a-Service (BaaS) was utilized for cloud-based database management, authentication, and secure storage of disaster-related data. The researchers also created system diagrams, dashboards, hazard maps, and responsive report forms to improve usability and situational awareness.

Daily Scrum. Throughout the development process, the researchers conducted regular discussions and progress monitoring to review completed tasks, identify technical issues, and determine the next development activities. This process helped ensure continuous collaboration, faster problem-solving, and proper integration of all modules within the system.

Sprint Review. After every sprint cycle, the developed features were presented and evaluated with Naic MDRRMO personnel. Functionalities such as hazard reporting, incident management, patient data collection, notifications, and dashboard analytics were tested to assess system usability, responsiveness, and effectiveness. Feedback gathered from users was used to improve and refine the system features.

Sprint Retrospective. At the end of each sprint, the researchers reviewed the overall development process to identify strengths, challenges, and possible improvements for succeeding sprint cycles. Adjustments and refinements were

implemented continuously to enhance the performance, security, and usability of the proposed SafeResponse system.

Deployment. After completing all sprint cycles, the proposed SafeResponse system was deployed as a functional web-based platform for Naic MDRRMO. The researchers conducted system demonstrations and orientations to familiarize MDRRMO personnel with the system features and operations. Final adjustments based on user feedback were also implemented to ensure the system's efficiency, usability, and reliability in supporting disaster response and hazard management operations.

System Testing

The proposed SafeResponse system was tested to ensure that all features and functionalities operated properly and without errors. The researchers tested the different modules, including hazard reporting, incident management, patient data collection, dashboard analytics, and geo-referenced hazard mapping, to verify that they functioned as intended. The system was also tested using sample data to evaluate its performance, responsiveness, and database synchronization. In addition, security testing was conducted to ensure that sensitive disaster-related information could only be accessed by authorized personnel. Alpha testing was performed by the researchers and selected Naic MDRRMO personnel to ensure proper system integration and usability.

System Evaluation

To prepare for the system evaluation, the researchers submitted a formal request letter to conduct the evaluation to the Adviser, Technical Critic, and concerned personnel of the Naic MDRRMO. Upon receiving approval, the

researchers utilized an evaluation form based on the ISO/IEC 25010 Software Quality Standards to assess the proposed system. The developed SafeResponse system was evaluated by IT Experts, MDRRMO Admin, MDRRMO Staff, and selected citizens of the Municipality of Naic. The researchers demonstrated the system to both technical and non-technical evaluators to assess its functionality, usability, performance efficiency, reliability, and security based on the ISO 25010 international software evaluation standard.

Participants of the study

Table 4 shows the participants of the study. The researchers aimed to evaluate the proposed SafeResponse system within the Naic MDRRMO and selected community users. The researchers utilized purposive sampling as the most appropriate sampling technique for selecting participants. The participants consisted of 2 IT Experts, 2 MDRRMO Admin personnel, 2 MDRRMO Staff, and 4 selected citizens, with a total of 10 participants involved in the system evaluation.

Table 4. Number of Participants

PARTICIPANTS	QUANTITY
IT Professionals	
MDRMMO Admin	2
MDRRMO Staff	2
Citizen/s	2
Total	10

Data analysis plan

Table 5 shows the descriptive interpretation of the mean which the numerical scale is starting from 1.50 and below will be interpreted as poor which is the lowest while the highest numerical scale of the mean is 4.51- 5.00 which will be interpreted as excellent. The corresponding interpretations were used in analyzing the result of the criteria.

Table 5. Descriptive interpretation of the mean

RANGE OF WEIGHTED MEAN	INTERPRETATION
4.51 - 5.00	Excellent
3.51 - 4.50	Very Good
2.51 - 3.50	Good
1.51 – 2.50	Fair
1.50 and below	Poor

Implementation Plan

The implementation plan outlines the process of deploying the proposed SafeResponse system to ensure effective use within the Naic MDRRMO. First is the Approval for Evaluation, where the researchers request approval from the MDRRMO Admin, Adviser, and Technical Critic to conduct the system evaluation. This phase may take 1–3 days and ensures that the system is ready for testing and assessment.

Next was Testing, where the researchers demonstrated the features and functionalities of the system to MDRRMO personnel and selected users. This 1–2

day process allows the users to explore the system, experience its functions, and provide feedback for possible improvements and adjustments.

Then comes System Evaluation, where evaluation forms are distributed to the researchers, MDRRMO Admin, staff, and selected users to assess the system's functionality, usability, and performance. This stage may take 1–5 days and is important in determining whether the system meets the needs and expectations of the users.

Finally, in the System Deployment, where the proposed SafeResponse system is officially deployed and introduced to the Naic MDRRMO. This phase may take 1–2 days and ensures that the system is fully prepared for actual use in disaster response and hazard management operations.

In conclusion, the implementation plan focuses on coordination, user feedback, and proper system preparation to ensure that the proposed SafeResponse system is functional, reliable, and effective in supporting disaster response and hazard management in the Municipality of Naic.

Table 6. Implementation plan

STRATEGY	ACTIVITIES	PERSONS INVOLVED	DURATION
Approval for evaluation	Requesting for approval of letter for the evaluation of the system	MDRRMO Admin, Staff, Technical Critic & Adviser	1- 3 days
Testing	Demonstrating the system to the Citizen, MDRRMO Admin, Staff	Researchers, Citizen, MDRRMO Admin, and Staff	1-2 days
System Evaluation	Evaluation of the system by providing the evaluation forms	Researchers, Citizen, MDRRMO Admin, and Staff	1-5 days
System Deployment	The researchers conduct an actual hands on to the user of the system	Researchers, MDRRMO Admin, and Staff	1-2 days